

Mr Tri Duc Nguyen
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Ultimo NSW 2007

MRA Automation
Sydney NSW Australia

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Dear MRA Automation team,

I am currently majoring in Mechatronics at the University Technology of Sydney. I learned of your company through offer of UTS Faculty of Engineering and Information Technology and I would like to apply for a Mechatronic engineering position. While I have a comprehensive electronic and mechanical background, my emphasis is on robotics and embedded system. Achieving a distinction of Mechatronics subject; along with experience about motor selection and control for my 2 own projects, I fairly believe that I can be hardworking with the job I am good at it.

Software Development, Algorithm Implementation, and Robust Testing

During a project simulating an autonomous drone maneuvering to a target floor within a complex structure in the Gazebo environment using ROS 2, we needed a reliable method to analyze system noise and vibration data. I was assigned with developing a data analysis function using Fast Fourier Transform (FFT) to process acquired test measurements and ensure the resulting analysis tool was robust and verifiable. I implemented the FFT algorithm within a Python function, designing it to efficiently process data separated into manageable chunks for reliable performance testing, and used Matplotlib to generate visualizations, including a required one-third octave plot for compliance checks against lab equipment standards. To guarantee reliability, I wrote comprehensive unit tests using the Pytest library, which were integrated into the project's version control workflow. The established FFT analysis function provided crucial frequency domain insights into the drone's control stability, enabling the engineering team to effectively tune the flight controllers, which led to a 15% reduction in positional error during simulated autonomous ascent.

Sensor Technologies, Sensor Fusion, and Data Analysis

Our autonomous industrial mobile equipment operated in environments prone to heavy dust and fog, where using only LiDAR resulted in unreliable obstacle detection and frequent false stops due to sensor noise. I was responsible for designing and implementing a robust, real-time sensor fusion module to combine data from the high-resolution LiDAR and the all-weather radar to create a reliable and continuous environmental model. In detail, I developed a fusion algorithm in C++ that used polar coordinate transformations to accurately map the radar's range and azimuth data onto the LiDAR's point cloud reference frame. Furthermore, I integrated an Extended Kalman Filter (EKF) to manage the temporal synchronization and uncertainty of the two disparate sensor streams, ensuring object tracking remained stable and precise even during visibility degradation. As a result, the fused perception system successfully filtered environmental noise, achieved a consistent real-time object tracking.

Upon graduating, I hope to work for a firm that be concerned with improving collaboration as well as project management skills. With the opportunities to work with professional team in real world industry, I could not only practice my technical skills, also enhance interpersonal skill so that I would advance my career.

Please let me know if you have any questions or would like to see some work samples. You can reach me by phone at (+61)415-957-947 or by email at Tri.NguyenDuc@student.uts.edu.au .

Yours sincerely,

Tri D Nguyen